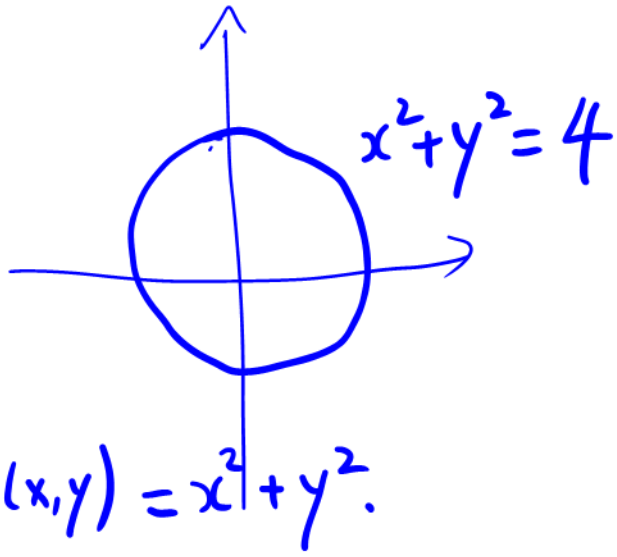
11. Keep the notation as in the previous question. Write down the system of three polynomial equations in the unkowns x, y and a parameter λ that you will need to solve if you use the method of Lagrange multipliers to find the maximum value of the function $f(x,y) = 2x^3 - 6x + 3xy^2$ on the circle $x^2 + y^2 = 4$.

$$\begin{cases} 6x^2 - 6 + 3y^2 = 2\lambda x \\ 6xy = 2\lambda y \\ x^2 + y^2 = 4 \end{cases}$$



Objective: To maximize

$f(x,y) = 2x^3 - 6x + 3xy^2$

subject to the constraint

$g(x,y) = 4$, where $g(x,y) = x^2 + y^2$.

For the method of Lagrange multipliers, we consider

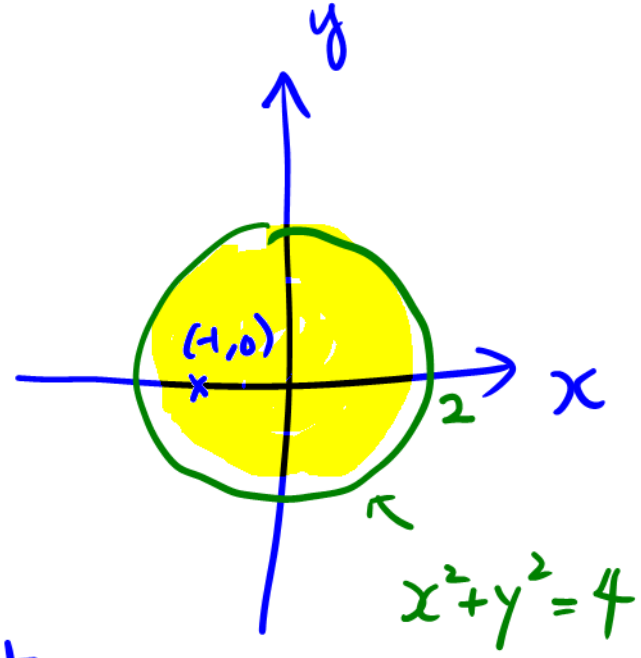
$$(*) \begin{cases} \nabla f(x,y) = \lambda \nabla g(x,y) \\ g(x,y) = 4. \end{cases}$$

$$\nabla f(x,y) = \begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} 6x^2 - 6 + 3y^2 \\ 6xy \end{pmatrix}, \quad \nabla g(x,y) = \begin{pmatrix} g_x \\ g_y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$(*) \iff \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ g(x,y) = 4 \end{cases} \iff \begin{cases} 6x^2 - 6 + 3y^2 = \lambda \cdot 2x \\ 6xy = \lambda \cdot 2y \\ x^2 + y^2 = 4 \end{cases}$$

12. Keep the notation as in the previous question. It is given that $(-1,0)$ is the only local maximum point of the function $f(x,y) = 2x^3 - 6x + 3xy^2$ in the xy-plane. Determine the maximum value of the function $f(x,y)$ on the closed disk of radius 2 centered at $(0,0)$.
Give the answer in the form $a\sqrt{b}$ where both a and b are positive integers and b is as small as possible.

$4\sqrt{2}$



Step 1: Look for candidate maximum point in the interior.
It is given that $(-1,0)$ is the only loc. max point of $f(x,y)$.

Step 2: Look for candidate maximum point along the boundary circle $x^2+y^2=4$.
From Q11, we solve

$$\begin{cases} 6x^2 - 6 + 3y^2 = 2\lambda x & \text{— (1)} \\ 6xy = 2\lambda y & \text{— (2)} \\ x^2 + y^2 = 4 & \text{— (3)} \end{cases}$$

From (2), $6xy = 2\lambda y \Rightarrow 2y(3x - \lambda) = 0 \Rightarrow y = 0$ or $\lambda = 3x$.

Case 1: $\lambda = 3x$. Substituting into (1),

$$6x^2 - 6 + 3y^2 = 2x \cdot 3x = 6x^2$$

$$\Rightarrow \quad 3y^2 = 6 \Rightarrow y^2 = 2 \Rightarrow y = \pm\sqrt{2}$$

Sub. $y = \pm\sqrt{2}$ into (3), $x^2 + 2 = 4 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$

$\therefore (x, y) = (\sqrt{2}, \sqrt{2}), (\sqrt{2}, -\sqrt{2}), (-\sqrt{2}, \sqrt{2}), (-\sqrt{2}, -\sqrt{2})$.

Case 2: $y = 0$. Subs. into (3), we get

$$x^2 + 0 = 4 \Rightarrow x = 2, -2$$

$\therefore (x, y) = (2, 0), (-2, 0)$

Step 3: Compare $f(x,y)$ at candidate points.

$$f(x,y) = 2x^3 - 6x + 3xy^2$$

$$f(-1,0) = -2 + 6 = 4$$

$$f(\sqrt{2}, \pm\sqrt{2}) = 2\cdot(\sqrt{2})^3 - 6\sqrt{2} + 3(\sqrt{2})(\pm\sqrt{2})^2$$

$$= 4\sqrt{2} - 6\sqrt{2} + 6\sqrt{2} = 4\sqrt{2}$$

$$f(-\sqrt{2}, \pm\sqrt{2}) = 2\cdot(-\sqrt{2})^3 - 6(-\sqrt{2}) + 3(-\sqrt{2})\underbrace{(\pm\sqrt{2})^2}_{2}$$

$$= -4\sqrt{2} + 6\sqrt{2} - 6\sqrt{2}$$

$$= -4\sqrt{2}$$

$$f(2,0) = 2\cdot 2^3 - 6\cdot 2 + 0 = 4$$

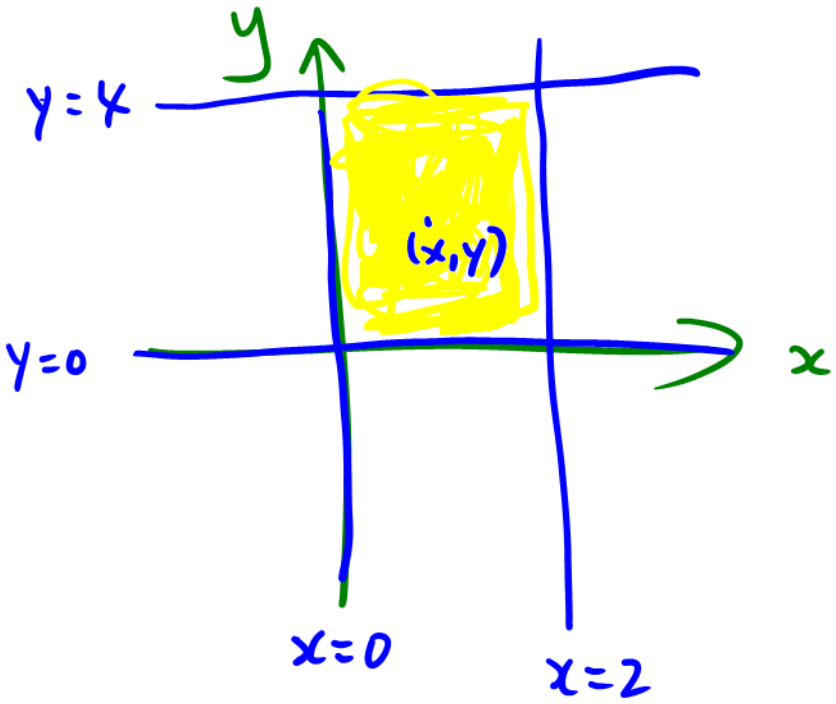
$$f(-2,0) = 2(-2)^3 - 6(-2) + 0 = -4$$

$\therefore$ max. value of $f(x,y)$ is $4\sqrt{2}$,
and is attained at $(\sqrt{2}, \sqrt{2})$ and $(\sqrt{2}, -\sqrt{2})$.

13. Let R be the region in the xy-plane bounded by the lines $x = 0$, $x = 2$, $y = 0$, $y = 4$. Evaluate the double integral $\iint_R (3x^2 + 4xy)\,dA$.

Give the answer as an integer or a fraction in lowest terms.

96



$$R = \{(x,y) : 0 \le x \le 2,\ 0 \le y \le 4\}$$

$$\iint_R (3x^2 + 4xy)\,dA$$

$$= \int_{x=0}^{x=2} \left(\int_{y=0}^{y=4} (3x^2 + 4xy)\,dy \right) dx$$

$$= \int_0^2 \left[3x^2 y + 2xy^2 \right]_{y=0}^{y=4} dx$$

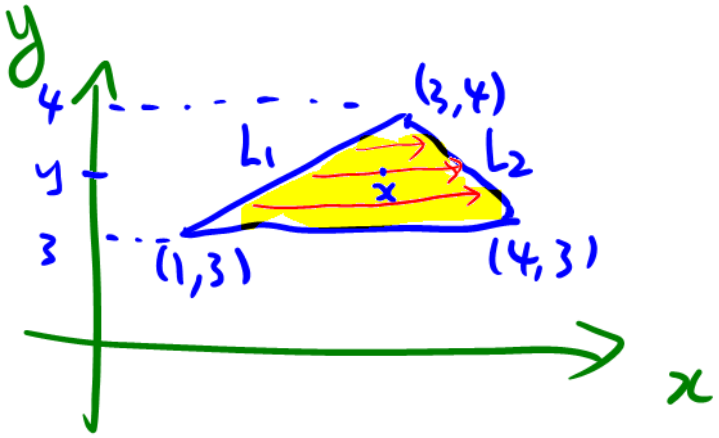
$$= \int_0^2 \left((12x^2 + 32x) - (0 + 0) \right) dx$$

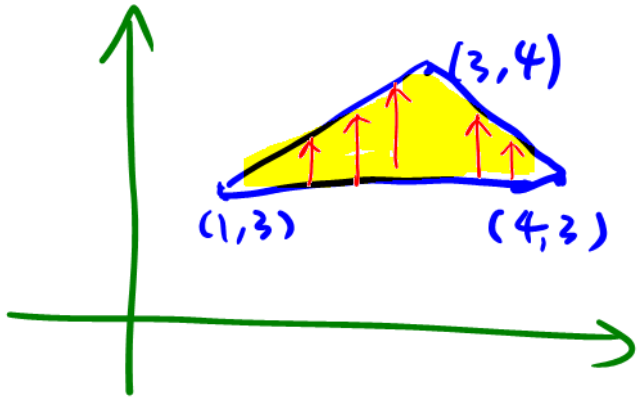
$$= \left[4x^3 + 16x^2 \right]_0^2 = 32 + 64 - (0+0) = 96 .$$

14. Consider the triangular region R in the xy-plane with vertices at $(1,3)$, $(4,3)$ and $(3,4)$. Compute $\iint_R y^2\,dA$.

Give the answer as an integer or a fraction in lowest terms.

$\frac{67}{4}$





Vertical arrow heads lie on 2 diff. line segments.

inner variable : x

outer variable : y

$$I = \iint_R y^2\,dA = \int_{y=3}^{y=4}\int_{x=2y-5}^{x=7-y} y^2\,dx\,dy$$

$$R = \{(x,y):\ 3 \le y \le 4,\ 2y-5 \le x \le 7-y\}$$

L_1 : $(1,3), (3,4)$

$$\frac{y-3}{x-1} = \frac{4-3}{3-1} = \frac{1}{2} \Rightarrow x-1 = 2(y-3) \Rightarrow x = 2y-5$$

L_2 : $(4,3), (3,4)$

$$\frac{y-3}{x-4} = \frac{4-3}{3-4} = -1 \Rightarrow x-4 = -(y-3) \Rightarrow x = 7-y.$$

$$I = \int_{y=3}^{y=4} \left(\int_{x=2y-5}^{x=7-y} y^2 \, dx \right) dy$$

$$= \int_{y=3}^{y=4} \left[y^2 x \right]_{x=2y-5}^{x=7-y} dy$$

$$= \int_{y=3}^{y=4} y^2 (7-y-(2y-5)) \, dy$$

$$= \int_{y=3}^{y=4} (12y^2 - 3y^3) \, dy$$

$$= \left[4y^3 - \frac{3}{4}y^4 \right]_3^4$$

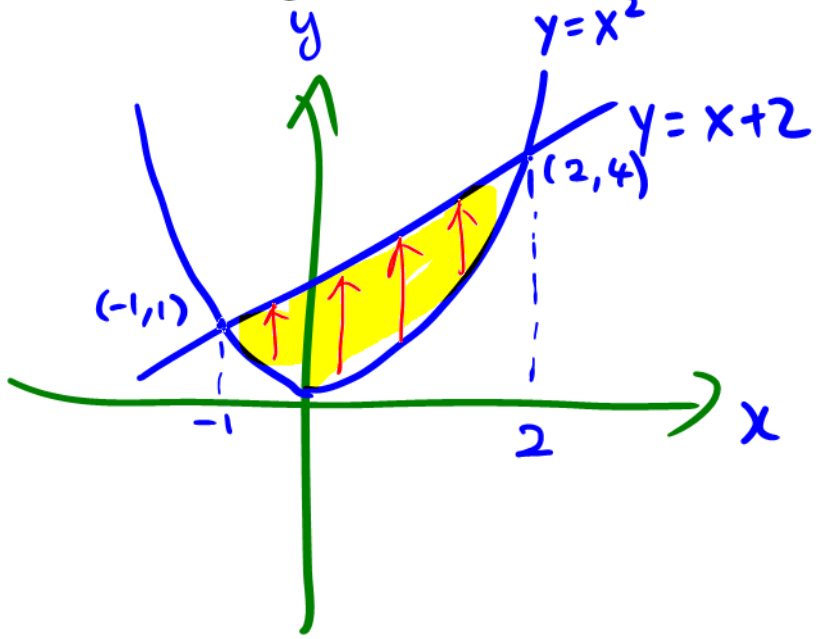
$$= (256 - 192) - (108 - \frac{243}{4})$$

$$= -44 + \frac{243}{4} = \frac{67}{4}.$$

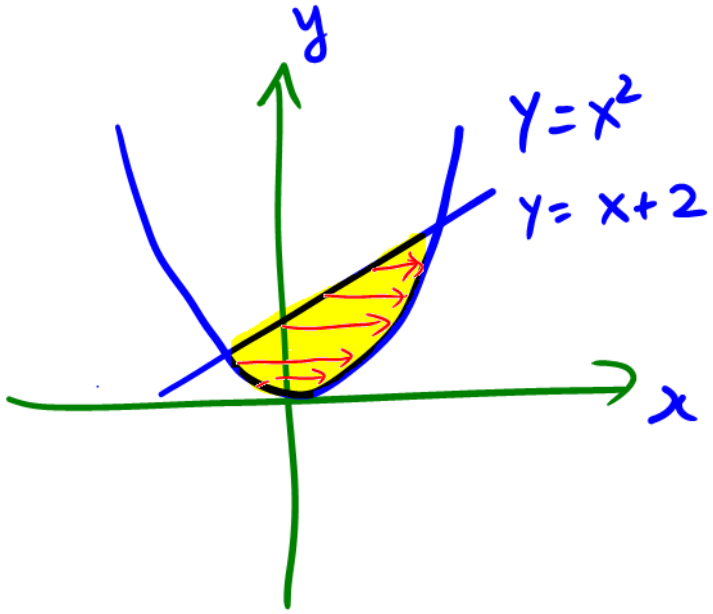
15. Let R be the region in the xy-plane bounded by the curve $y = x^2$ and the line $y = x + 2$. Find the value of the double integral $\iint_R x\,dA$.

Give the answer as an integer or a fraction in lowest terms.

$\frac{9}{4}$



inner variable: y
outer variable: x



Tails of horizontal arrow lie on 2 diff. curve/line.

To find the points of intersection, we solve

$$\begin{cases} y = x^2 & -(1) \\ y = x+2 & -(2) \end{cases}$$

Sub. (2) into (1), we get $x^2 = x+2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0$

$\Rightarrow x = 2, -1$

$R = \{(x,y) : -1 \le x \le 2,\ x^2 \le y \le x+2\}$ $\Rightarrow (x,y) = (2,4), (-1,1)$.

$$\iint_R x \, dA = \int_{x=-1}^{x=2} \left(\int_{y=x^2}^{y=x+2} x \, dy \right) dx$$

$$= \int_{x=-1}^{x=2} \Big[xy \Big]_{y=x^2}^{y=x+2} dx$$

$$= \int_{-1}^{2} \left(x(x+2) - x \cdot x^2 \right) dx$$

$$= \int_{-1}^{2} \left(x^2 + 2x - x^3 \right) dx$$

$$= \left[\frac{x^3}{3} + x^2 - \frac{x^4}{4} \right]_{-1}^{2} = \left(\frac{8}{3} + 4 - 4 \right) - \left(-\frac{1}{3} + 1 - \frac{1}{4} \right)$$

$$= \frac{9}{4}.$$

16. Find the exact value of the iterated integral

$$I = \int_0^1 \int_{x^{1/3}}^1 \frac{1}{\sqrt{y^4+1}}\, dy\, dx.$$

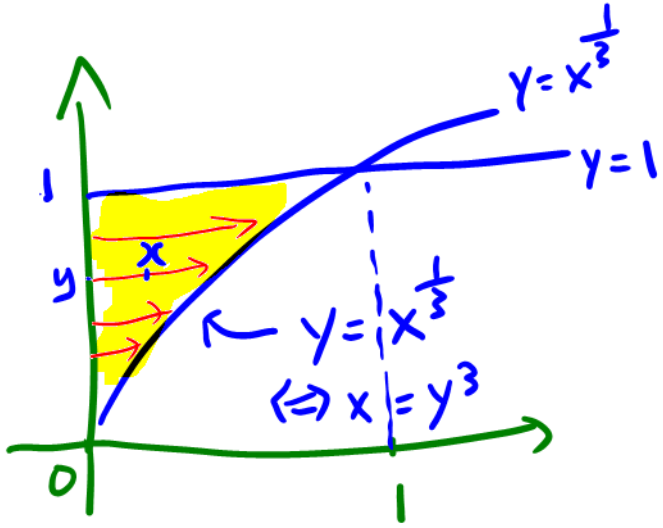
$\frac{\sqrt{2}-1}{2}$.

$\int \frac{1}{\sqrt{y^4+1}}\, dy$ is difficult to compute.

So try reversing order of integration.

$$R = \{(x,y) : 0 \le x \le 1,\ x^{\frac{1}{3}} \le y \le 1\}$$

$$= \{(x,y) : 0 \le y \le 1,\ 0 \le x \le y^3\}$$



$$I = \int_{y=0}^{y=1} \left(\int_{x=0}^{x=y^3} \frac{1}{\sqrt{y^4+1}}\, dx \right) dy$$

$$= \int_{y=0}^{y=1} \left[\frac{x}{\sqrt{y^4+1}} \right]_{x=0}^{x=y^3} dy$$

$$= \int_{y=0}^{y=1} \left(\frac{y^3}{\sqrt{y^4+1}} - 0 \right) dy$$

$$= \left[\frac{1}{2}\sqrt{y^4+1} \right]_{y=0}^{y=1}$$

$$= \frac{\sqrt{2}-1}{2}$$

Let $u = y^4+1$

$$\frac{du}{dy} = 4y^3$$

$$y^3 dy = \frac{du}{4}$$

$$\int \frac{y^3}{\sqrt{y^4+1}} dy = \frac{1}{4}\int \frac{1}{\sqrt{u}} du$$

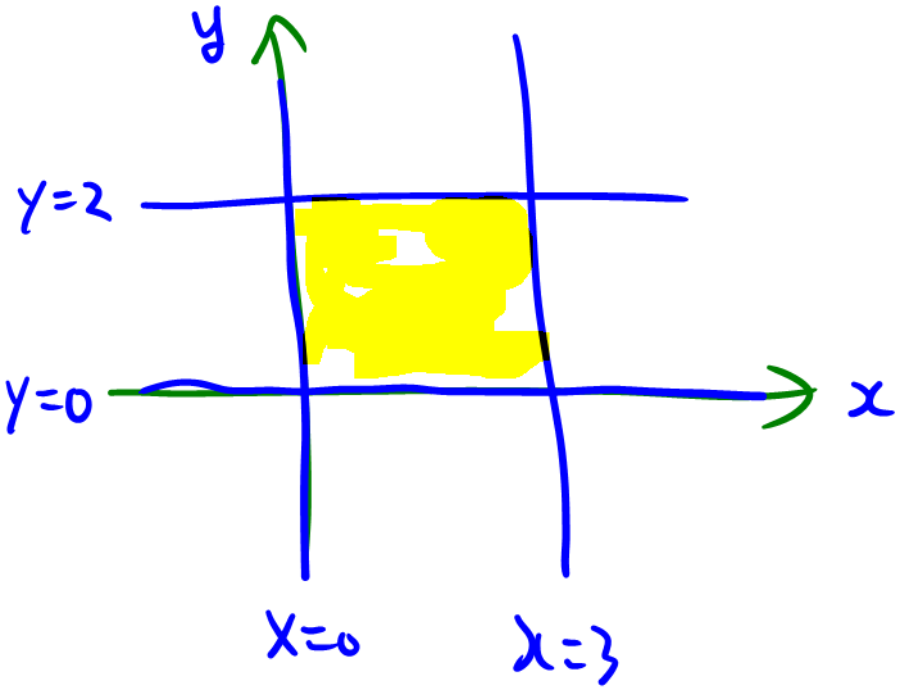
$$= \frac{1}{4} \cdot \frac{u^{\frac{1}{2}}}{\frac{1}{2}}$$

$$= \frac{1}{2}\sqrt{y^4+1} + C$$

17. Let $f(t)$ be a function such that $\int_0^6 f(t)\,dt = 18$. Let R the region in the xy-plane bounded by the lines $x = 0$, $x = 3$, $y = 0$ and $y = 2$. Find the value of the double integral $\iint_R f(2x)f(3y)\,dA$.

Give the answer as an integer or a fraction in lowest terms.

54



$R = \{(x,y) : 0 \le x \le 3,\ 0 \le y \le 2\}$

$$I = \iint_R f(2x)f(3y)\,dA$$

$$= \int_{y=0}^{y=2} \left(\int_{x=0}^{x=3} f(2x)f(3y)\,dx \right) dy$$

$$= \int_{y=0}^{y=2} f(3y) \int_{x=0}^{x=3} f(2x)\,dx\ dy$$

$$= \left(\int_{x=0}^{x=3} f(2x)\,dx\right)\cdot\left(\int_{y=0}^{y=2} f(3y)\,dy\right)$$

Given: $\int_0^6 f(t)\,dt = 18.$

For $\int_{x=0}^{x=3} f(2x)\,dx$, put $u = 2x$

$\frac{du}{dx} = 2$, $dx = \frac{du}{2}$

When $x=0$, $u=0$

$x=3$ $u=6$.

$$\int_{x=0}^{x=3} f(2x)\,dx = \int_{u=0}^{u=6} f(u)\,\frac{du}{2} = \frac{1}{2}\int_{u=0}^{u=6} f(u)\,du = \frac{1}{2}\cdot 18 = 9.$$

$$\int_{y=0}^{y=2} f(3y)\,dy$$

Put $v = 3y$, $\frac{dv}{dy} = 3 \Rightarrow dy = \frac{dv}{3}$

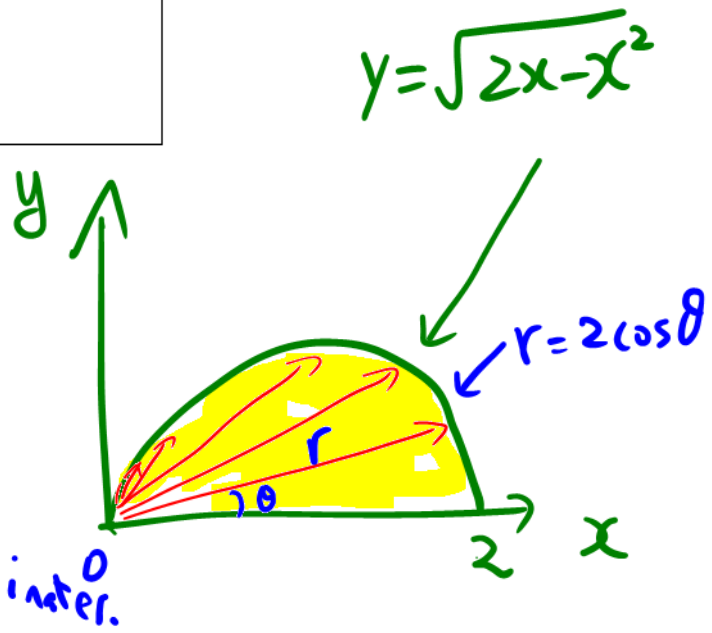
When $y=0$, $v=0$; when $y=2$, $v=6$

$$= \int_{v=0}^{v=6} f(v)\,\frac{dv}{3} = \frac{1}{3}\int_{v=0}^{v=6} f(v)\,dv = \frac{1}{3}\cdot 18 = 6.$$

$\therefore \quad I = 9\cdot 6 = 54.$

18. Consider the region R in the first quadrant of the xy-plane bounded by the curve $y = \sqrt{2x - x^2}$ and the x-axis. Find the value of the double integral $\iint_R \frac{1}{x}\, dA$. Give the answer in terms of π.

π.



$R = \{(x,y) : 0 \le x \le 2,\ 0 \le y \le \sqrt{2x-x^2}\}$

Remark: Difficult to compute $\iint_R \frac{1}{x}\, dA$ using Cartesian coordinates.

Use polar coordinates (r,θ): $x = r\cos\theta$, $y = r\sin\theta$, $dA = r\, dr\, d\theta$

$R = \{(r,\theta) : 0 \le \theta \le \frac{\pi}{2},\ 0 \le r \le 2\cos\theta\}$

For $r > 0$, $y = \sqrt{2x-x^2} \Rightarrow y^2 + x^2 = 2x \Rightarrow r^2 = 2r\cos\theta$

$\Rightarrow r = 2\cos\theta$.

$$\iint_R \frac{1}{x}\,dA = \int_{\theta=0}^{\theta=\frac{\pi}{2}} \left(\int_{r=0}^{r=2\cos\theta} \frac{1}{r\cos\theta}\, r\, dr \right) d\theta$$

$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \left[\frac{r}{\cos\theta} \right]_{r=0}^{r=2\cos\theta} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{2\cos\theta - 0}{\cos\theta}\, d\theta$$

$$= \left[2\theta \right]_0^{\frac{\pi}{2}} = \pi.$$